Higher National Diploma in Quantity Surveying

Second Year, 1st Semester Examination – 2017

QS 21054- Civil Engineering Construction ii for QS

Instructions for candidates:

Answer five Questions out of six
All questions carry equal marks.
Calculators are allowed.

No of pages : 06
No of questions: 06
Time: Three (03) hours

Q (01) Cement is a useful binding material in the concrete work. Quality controlling of cement manufacturing is important to get the required strength of concrete.

(a) State the two process of manufacturing cement. (02)

(b) State the 04 important properties of cement. (04)

(c) What type of cement do you propose to construction in costal belt area? (02)

(d) If it is want to remove the formwork in early what type of cement are you recommend (02)

(e) Give the reason of storing cement in timber flat form with air tight (02)

(f) Draw a line diagram with name of the stages and row material to present manufacturing process of cement (08)

Q (02) Two essential properties of fresh concrete are workability and compatibility and two essential properties of hardened concrete are crushing strength and durability.

(a) Explain the two test used in field to measure the workability. (06)

(b) What type of defects can be occurred in placing of fresh concrete? (03)

(c) External and internal vibrations are needed for proper compaction of fresh concrete Explain briefly external and internal vibration (03)

(d) Ready mix concrete is popular for large scale concrete work. What are transporting and pouring method you plan for the work. (03)

(e) What the purposes of use admixtures in concrete? (03)

(f) Briefly explain the control of heat generated in hydration action in hardening stage of concrete. (02)

Q (03) Design of Multi story building and non-Domestic building, Reinforce Cement Concrete (RCC) are used in substructure and super structure. During the cost estimating stages of a proposed project Quantity Surveys are concern about the cost of a steel building.

(a) Explain briefly what the components in multi-story building designed with RCC (03)
(b) What are the disadvantages of steel framed structures used multistory building? (03)
(c) What are the loads are you considering in design of RCC frame Structure? (04)
(d) Briefly explain the elements of substructure and super Structure of a Building (03)
(e) State three possible failure modes in RCC structures. (03)
(f) What are the standards and specifications should be considered for structural application of Steel for Reinforcement as well as steel Column and Beam sections. (02)
(g) What are the most common joints applicable in steel frame structures (02)

Q (04) it is difficult to maintain the constant strength of concrete in production practice due to the various factors of aggregates. Therefore it is needed to carry out the Mix Design to the required Characteristic Strength and workability. By considering probability theories Target mean strength (ft) to be calculated and do the mix design for that strength to Maintain required Characteristic Strength (fc)

(a) What are the factors of aggregate cause to reduce strength of concrete? (03)
(b) What do you mean by Target mean strength in mix design? (03) (c) Explain the volume batching and weigh batching in mix proportion of concrete (02)
(d) Why weigh batching is preferred in mix design calculation (02)
(e) Determine the quantities of cement, water, coarse aggregate and Fine aggregate to produce 10 m³ of concrete with relative to following data. Relevant charts and tables are attached here with.

- Characteristic Strength specified in concrete is 45 N/ mm² at 28 days.
- K= 1.96 for 2.5%defects.
- Samples tested more than 20.
- Cement strength Class 42.5
- slump is 30-60 mm,
- maximum aggregate Size is 20 mm,

- Aggregate type coarse -crushed and fine - uncrushed.
- Specific gravity of Aggregate is 2.7
- aggregate: 70% passing 600 μm sieve (uncrushed)

(f) Calculate the Target mean strength of the concrete is at 28 days (03)

(g). Determine the quantities of cement, water, coarse aggregate and fine aggregate for 10 m^3 (07)

Q (05) Building services are highly important for commercial building than Domestic building. Multi story Commercial buildings and Apartments are much concentrated on building services because of the comfort and healthy of community.

(a) What are traditional and modern building services are needed for Commercial buildings and Apartments? (02)

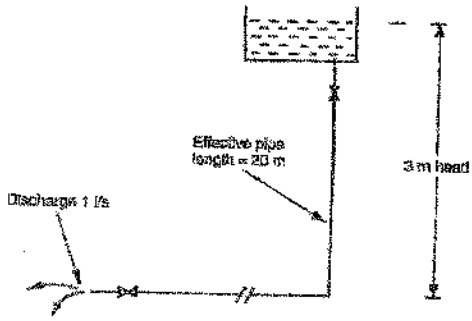
(b) Waste water and Storm water Management are crucial in urban areas. What are the steps to be implemented to minimize the situation? (02)

(c) Design the diameter of water service line for domestic water distribution system shown below. Use the **Thomas Box Formula**. Assume no head losses due to bends tees and specials etc. (04)

Thomas Box formula:

$$d = \sqrt[5]{\frac{q^2 \times 25 \times L \times 10^3}{H}}$$

where: d = diameter (bore) of pipe (mm)
q = flow rate (l/s)
H = head or pressure (m)
L = length (effective) of pipe (m)
(actual length + allowance for bends, tees, etc.)



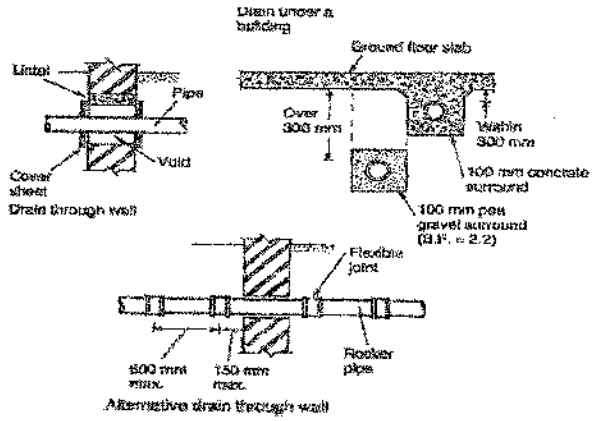
(d) What are the pre service arrangements can be provided to evacuate people from sudden fire in commercial building? (02)

(e) Heating Ventilation and Air-condition (HVAC) are services equipped with electrical and mechanical equipment. How can implement Green Energy concept for HVAC to serve the energy consumption. (02)

(f) Specify the major concerns when design of Lift Installations in multistory building (06)

(g) What do you mean by Conduits and Service Ducts in a Building (02)

(h) Explain the locations in building where we are using this service arrangement (02)



(f) Draw a Sketch diagram by name all parts for domestic septic tank and Soakage Pit (06)

Q (06) briefly explain following

(a) Dead Loads, Live Loads and Wind Loads (02)
(b) Precast Concrete (02)
(c) Non load bearing walls and shear walls (02)
(d) Pre Tension and Post Tension of precast concrete (02)
(e) Formwork, Scaffolding and Shoring (06)
(f) Draw a shoring arrangement for trench excavation with name all components (06)

Table 2 Approximate compressive strength (N/mm²) of concrete made with a free-water/cement ratio of 0.5

Cement strength class	Type of coarse aggregate	Compressive strength (N/mm²) Age (days) 3	7	28	91
42.5	Uncrushed	22	30	42	48
	Crushed	27	36	49	55
52.5	Uncrushed	29	37	49	54
	Crushed	34	43	55	61

Increase the tabulated compressive strengths by 10 N/mm² (1 kbar) for each 0.1 increase in cement content above 300 kg/m³.

Table 3 Approximate free-water contents (kg/m³) to give a range of workability

Slump (mm)		0–10	10–30	30–60	60–180
Vebe time (s)		>12	6–12	3–6	0–3
Maximum size of aggregate (mm)	Type of aggregate				
10	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

Note: When curves for aggregates of different types are used, the free-water content is corrected by the expression:

$$w_f = w_{f_u} + w_{f_c}$$

where w_{f_u} = free-water content appropriate to type of fine aggregate

and w_{f_c} = free-water content appropriate to type of coarse aggregate

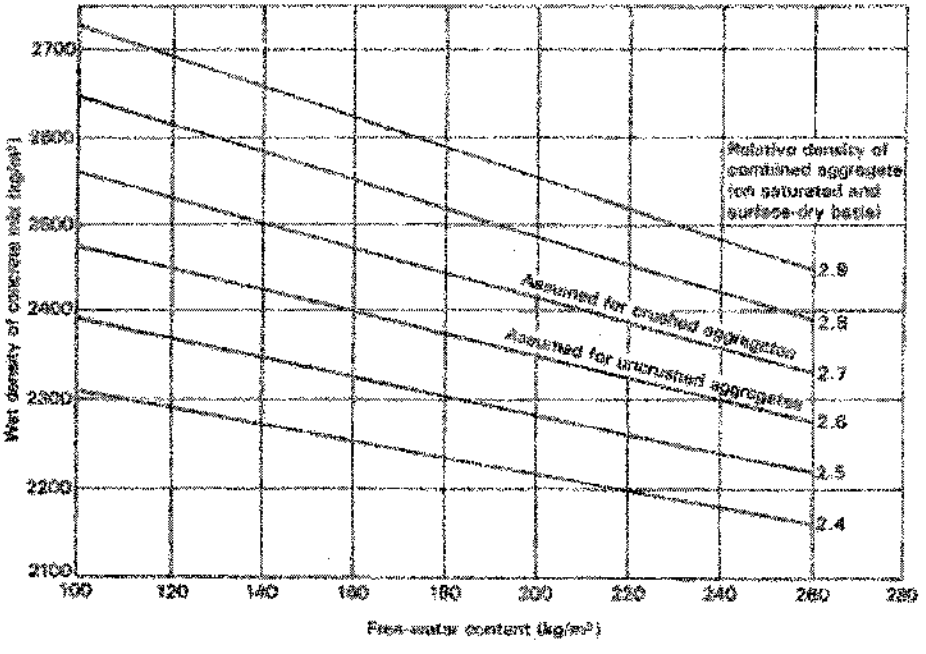
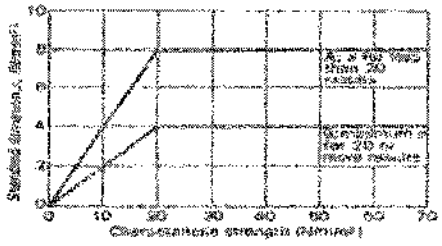
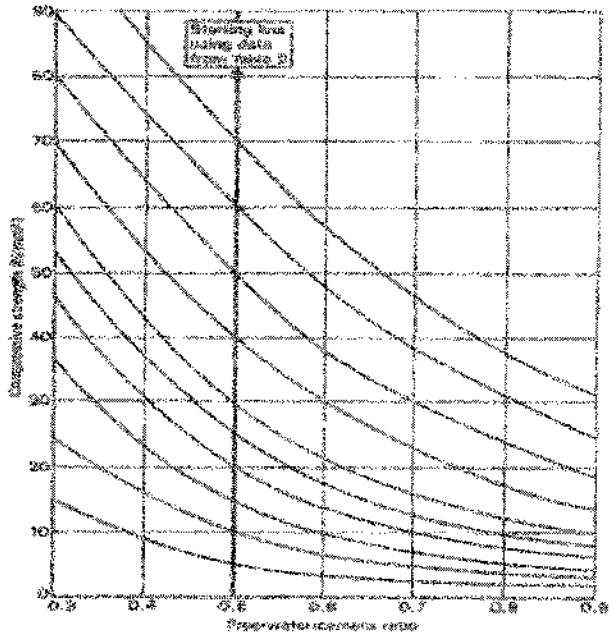
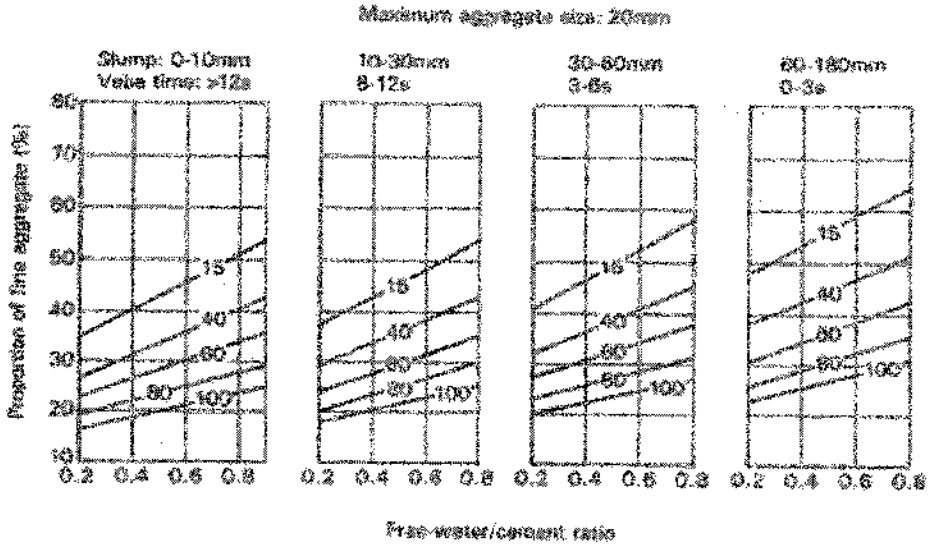


Figure 6 Estimated wet density of fully compacted concrete

Tables and Graphs related to Q4